Kennesaw State University

College of Computing and Software Engineering

DEPARTMENT OF COMPUTER SCIENCE

CS4412 – Data Mining – Section W01

Data Mining Project

M2 – Initial Implementation

Author
By Cesar Arevalo Colocho
careval3@students.kennesaw.edu

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Student Performance Pattern Mining Using Clustering Analysis

1. Introduction

This paper examines behavioral and academic performance trends of the students based upon the Student Performance dataset, which was acquired in the UCI Machine Learning Repository. The data set consists of demographic, social, lifestyle, and academic performance variables obtained from the students that are registered in Mathematics and Portuguese classes in secondary schools.

The major aim of this analysis is to use the exploratory data mining methods to discover the natural fraction of students using indicators of academic activity. Pattern discovery and behavioral segmentation are more emphasized in this project instead of predictive modeling. Three discovery questions guided the analysis:

1. Do family and socioeconomic attributes form recognizable patterns associated with academic consistency?
2. How do study habits and absence behavior interact with student performance?
3. What natural clusters of students emerge when combining academic, demographic, and lifestyle factors?

2. Methodology

The analysis followed a structured data mining workflow (figure 1) consisting of data exploration, preprocessing, and clustering-based pattern discovery.

Exploratory Data Analysis

The preliminary analysis of the dataset proved the hypothesis that the dataset has 1044 observations and 34 attributes. Each record was found to be unique and not a duplicate, meaning that there were no missing values or records to be used in the data mining.

- **Student profile overview**

The analysis of distribution indicated that a majority of the students belonged to the age bracket of 15-18 years, which is the normal age group in secondary schools. There was a relatively equal number of girls and boys, and the number of female students outnumbered the number of male pupils, and most students live in urban areas. Family related plots show that most students live with both parents, many belong to larger families, and family support is common. In contrast, only a smaller share of students attend paid extra classes.

- **Academic performance trends**

Grade distributions across G1, G2, and G3 are constant. The medians remain similar across the three evaluation periods, and the spread is also comparable, with slightly greater variation in the final grade. This consistency suggests that early academic performance is a strong signal of later performance rather than a random short-term fluctuation.

- **Study habits, absences, and failures**

Study time shows a modest upward relationship with final grade: students in higher study-time categories tend to achieve somewhat better G3 scores, although the overlap between groups remains substantial. Absences show only a weak negative relationship with final grade. Previous failures, however, display a much clearer negative pattern, making failure history a stronger warning sign than attendance alone.

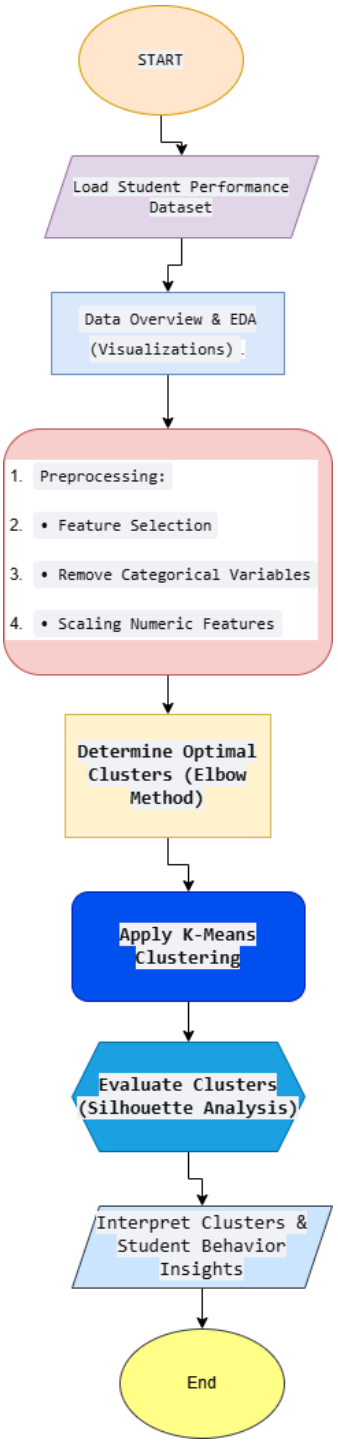

Figure 1 - Workflow

- **Socioeconomic and family factors**

Family relationship quality is generally positive across the sample, with most students reporting good or excellent family relationships. Mother's and father's education both show slight upward trends in relation to final grade, suggesting a modest parental education effect. Paid extra classes do not clearly separate high performing from low performing students, and internet access appears beneficial but not dominant as a stand alone predictor.

3. Data Preprocessing

Preprocessing was done in order to make it compatible with clustering algorithms. Several binary yes/no variables were converted to 0/1 format. Other nominal fields such as sex, address, family size, parental status, school, job type, reason, guardian, and course were also encoded numerically for general preprocessing and visualization. However, the clustering model deliberately avoids using arbitrary numeric codings for purely nominal categories because Euclidean distance can create misleading similarities between categories.

For K-means clustering, the selected variables were scaled after feature selection. The feature set used in clustering consists of studytime, failures, absences, G1, G2, G3, Medu, and Fedu. These variables were chosen because they capture study behavior, academic history, attendance, and parental education without overloading the model with weakly interpretable categorical encodings

4. Correlation Analysis

The correlation matrix confirms that academic history variables dominate the signal in the dataset. G1 is strongly correlated with G2 (about 0.86) and G3 (about 0.81), while G2 is very strongly correlated with G3 (about 0.91). These are the strongest relationships observed and show that earlier grades are the best indicators of later grades.

Failures have a moderate negative relationship with grades (approximately -0.37 to -0.38).

Study time has only a weak positive correlation with grade outcomes, and absences show a weak negative relationship. Family support and family relationship measures contribute useful context, but their direct statistical association with final performance is much weaker than prior grade information.

Relationship	Approx. value	Interpretation
G1 ↔ G2	0.86	Strong positive relationship
G1 ↔ G3	0.81	Strong positive relationship
G2 ↔ G3	0.91	Very strong positive relationship
Failures ↔ G3	-0.38	Moderate negative relationship
Study time ↔ G3	0.16	Weak positive relationship
Absences ↔ G3	-0.05	Weak negative relationship

5. Clustering Method and Evaluation

K-means clustering was selected as the initial pattern-mining technique because it provides an accessible first segmentation of student profiles. The elbow method was used to select the number of clusters, and k = 3 was chosen because the within-cluster sum of squares begins to level off after three clusters.

The resulting cluster solution is interpretable, but its separation is only moderate. The overall average silhouette score is approximately 0.19, which means the clusters are present but not sharply divided. Cluster level silhouette values indicate that one group is much less distinct than the others, so the model is useful as an exploratory starting point rather than a final segmentation answer.

6. Cluster Interpretation

Cluster	Profile	Main characteristics
Cluster 1	Higher-performing students	Stronger G1–G3 performance, very low failure history, and generally favorable academic profile.
Cluster 2	Moderate-performing students	Average grades, relatively low failures, and middle-range behavior across study and attendance indicators.
Cluster 3	Academically at-risk students	Lowest average grades, highest average failures, and highest average absences among the three groups.

This three group structure supports the idea that student performance is not random; instead, students cluster into broad profiles that reflect combinations of prior grades, study behavior, and academic risk. At the same time, the modest silhouette value suggests the boundaries between these groups are somewhat fuzzy, which is very normal in education data, where humans stubbornly refuse to organize themselves into perfectly neat circles for K-means.

7. Mining Technique: K-Means Clustering

The major pattern discovery technique that was chosen was K-means clustering because it is effective in identifying small groups of student behavior.

The number of clusters was selected optimally through the elbow method of assessment that examines within-cluster sum of squares at various levels of clusters. Judging by the look of the elbow curve, three clusters were chosen as the optimal segmentation point of this data.

Stated behavioral and academic attributes were used to train the clustering model. Each student observation was then assigned cluster labels, enabling a statistical meaning at a group level.

Also done was the silhouette analysis to determine the quality of the clusters. The silhouette plot showed the moderate level of coherence in clusters, which is the reason to believe that the chosen feature set will offer significant behavioral segmentation.

8. Pattern Discovery Results

The analysis of the cluster profile indicated that there were specific segments of student behavior.

The former group depicts students that are relatively high achieving. Students who belonged to such a group tended to have a moderate to high study time, reduced instances of absenteeism, and high-grade historical performance not only in G1 but also in G2 and G3.

The second group seems to be a case of academically vulnerable students. This group was described as having reduced levels of study engagement, an escalated history of failure, and an augmented frequency of absenteeism. There are behavioral signs that indicate that attendance and evening study habits would be significant determinants of academic success.

The third cluster is representative of ambivalent behavior, moderate academic achievement, and inconsistent lifestyle and social activities.

Overall, historical academic performance variables proved to be the largest contributors to cluster separation, especially the sequential grade measures.

9. Answers to Discovery Questions

The analysis gives information about the initial research questions.

The secondary effect was demonstrated by family and socioeconomic factors over direct indicators of academic behavior. Other variables that assisted in clustering structures were family relationship quality and health status but were not the overriding segmentation drivers.

Academic outcomes had an important correlation with study habits. Students who had more time studying and a low failure history were more likely to be in the high-performance group. The clustering process was able to isolate natural groups of students in behavior, and this indicates academic performance can be partially explained by behavioral segmentation.

- **Family and Socioeconomic patterns**

Yes, family and socioeconomic variables do show recognizable patterns, but their influence is generally modest. Positive family relationships are common, and parental education has a slight positive association with academic performance. Paid extra classes and internet access provide some context, but they are not the primary drivers of academic consistency.

- **Study habits and Absences**

Study habits and absences matter, but not equally. More study time is associated with somewhat better grades, while absences show only a weak negative trend. Previous failures are much more informative than absences, indicating that longer-term academic history is a stronger risk marker than attendance by itself.

- **Natural Student Clusters**

Three broad student clusters emerge: higher-performing, moderate-performing, and academically at risk. These clusters are shaped more strongly by grade history and failures than by isolated socioeconomic variables. This means academic history should remain central in future interpretation and intervention strategies.

10. Technical Challenges and Considerations

A number of possible limitations were found in the analysis such as :

- The optimal number of clusters to use needed to be decided through an iterative process by the elbow method.
- There were also problems in feature selection since incorporating irrelevant variables can decrease the performance of the clustering.

- Managing behavioral noise in variables of student lifestyle is also one of the possible areas of improvement of another analysis.

11. Future Work for M3

Future research directions include:

- Introduction of hierarchical clustering techniques of hierarchy of student segmentation.
- Association rule mining: Apriori or FP-Growth exploration of behavioral rule discovery.
- Encoding of categorical features to include more demographic variables.
- Clustering stability validation with other measuring icons of distance.
- Academic performance prediction model development.

12. Conclusion

This first application was a clear cut illustration of how clustering-based methods of data mining techniques can be used in the discovery of student behavior patterns. The findings show that both the variables of academic history and study behavior are important in the creation of natural student groups.

Overall, the M2 work successfully identifies interpretable student profiles and sets up a clear path for deeper validation in later milestones. In plain terms: the project is on the right track, the data is behaving, and the students have kindly revealed just enough patterns to keep the next milestone interesting.

The analysis serves as a basis for the further development of pattern mining exploration in the further project milestones. Although in the present research, the emphasis is on the segmentation discovery, further research in the future will be directed towards the discovery of more profound rule mining and validation-based modeling methods.